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VII SEMESTER

Experiential Learning - ISM

Phase 1

on

DRIVE: Data-Driven Reliability Intelligence for Vulnerability Estimation

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Introduction

Managing Reliability Across Large SSD Fleets

- Modern data centres rely on large SSD populations to support cloud services, databases and critical applications.
- Drive failures can lead to service disruption, data unavailability and additional maintenance costs.
- Failure occurrences may vary across SSD models and infrastructure locations such as racks and nodes.
- Since inspecting or replacing every drive is impractical, maintenance teams must prioritise where resources are allocated.

Project Motivation

Use historical SSD failure patterns to identify drives and infrastructure groups that deserve maintenance attention first.

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Literature Review

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S.No	Title of Paper	Author(s)	Key Findings	Research Gaps (Considering Our Project)
1	An In-Depth Study of Correlated Failures in Production SSD-Based Data Centers (2021)	Shujie Han et al.	Analyses large-scale SSD failure records and identifies correlated failures across infrastructure locations.	Does not compare budget-constrained maintenance selection strategies using model-wise and infrastructure-wise failure patterns.
2	Prophet: SSD Failure Analysis and Prediction Guided by Flash Reliability Characteristics in Data Centers (2025)	Yunpeng Song, Yujiong Liang, Jialin Liu, Liang Shi	Incorporates flash-memory reliability characteristics into SSD failure analysis and prediction.	Does not focus on converting historical failure patterns into explainable maintenance priorities.
3	Temporal-Contextual Attention Network for Solid-State Drive Failure Prediction in Data Centers (2024)	Chankyu Koh, Jisu Kang, Taehyeong Kim, Sung Won Han	Uses LSTM and Transformer-based attention to capture temporal patterns and relationships among SMART attributes.	Focuses on prediction accuracy rather than resource-constrained maintenance selection and prioritisation.

Literature Review

S.No	Title of Paper	Author(s)	Key Findings	Research Gaps (Considering Our Project)
4	Multiple Instance Learning for SSD Failure Prediction Under Customer Failure-Biased Labels (2026)	Bongjun Choi, Jeongwon Park, Hyung-Seok Kang, Jaewoong Shim	Uses multiple instance learning and spatial-temporal information to improve prediction under noisy failure labels.	Does not investigate comparative maintenance prioritisation strategies under fixed resource budgets.
5	Estimating Mean Time to Failure of Solid-State Drives via Self-Organizing Map and Model Averaging (2025)	Peng Li, Xun Xiao, Jiayu Chen	Combines health-indicator construction and degradation modelling to estimate SSD mean time to failure.	Does not address fleet-level maintenance prioritisation using infrastructure-level failure patterns.
6	Online Maintenance Prioritization via Monte Carlo Tree Search and Case-Based Reasoning (2022)	Michael Hoffman, Eunhye Song, Michael Brundage, Soundar Kumara	Uses Monte Carlo Tree Search and case-based reasoning to prioritise maintenance under resource constraints.	Focuses on general maintenance decision-making rather than SSD-specific historical failure analysis and infrastructure-aware selection.

Problem Definition

Data centre operators manage large SSD populations with limited maintenance resources. Historical failure patterns vary across SSD models and infrastructure groups, making uniform or random maintenance selection potentially inefficient.

A systematic approach is required to identify and prioritise SSDs for maintenance while maximising the coverage of historically observed failures.

Objectives

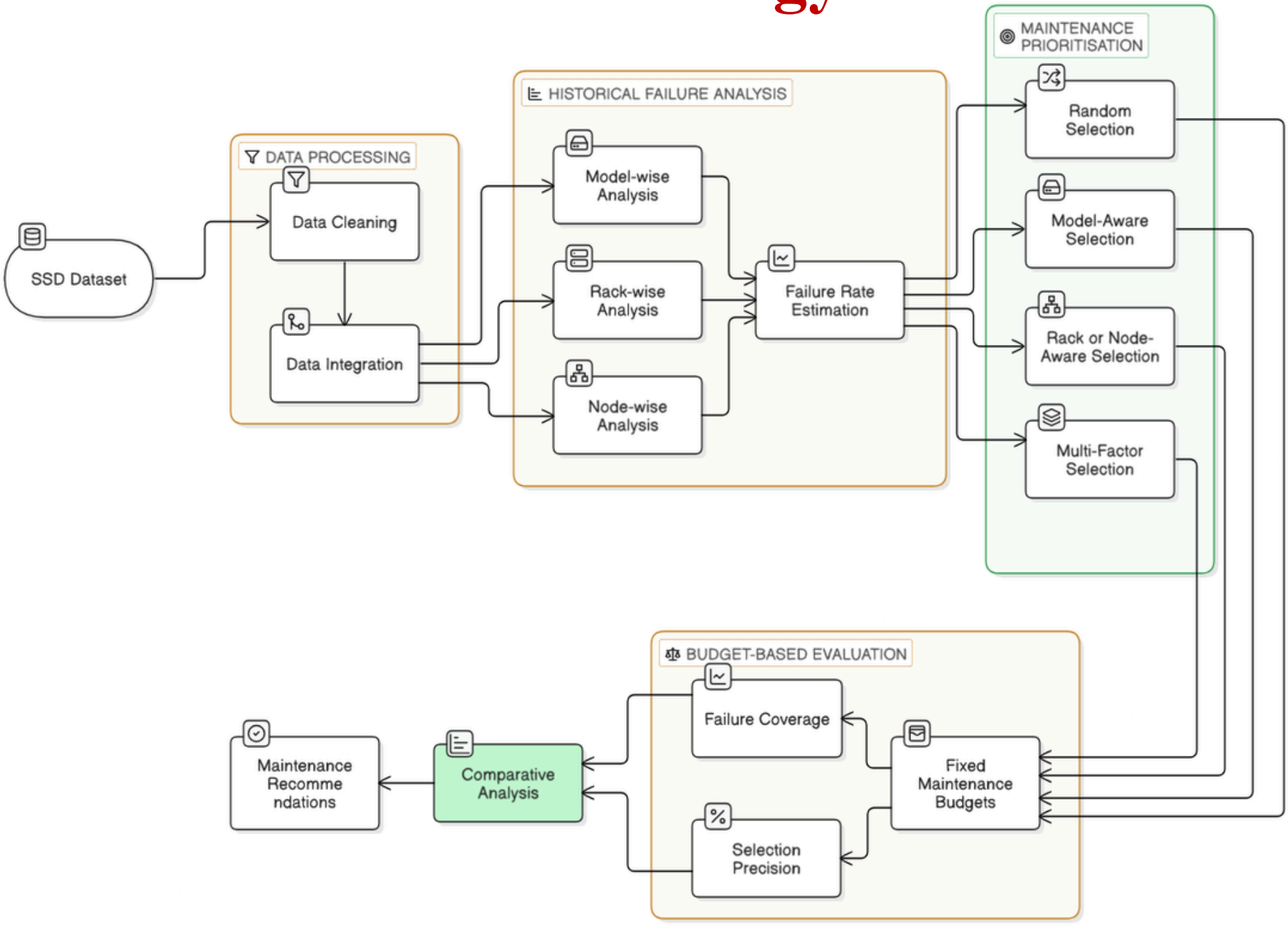
Primary Objectives:

- To analyse historical SSD failure patterns across SSD models and infrastructure groups.
- To develop explainable, failure-aware maintenance prioritisation strategies using historical failure distributions.
- To evaluate and compare maintenance strategies under different maintenance budget constraints.

Secondary Objectives:

- To visualise SSD failure patterns, maintenance priorities and strategy performance through an interactive dashboard.
- To provide interpretable recommendations for efficient allocation of limited maintenance resources.

Methodology



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Dataset Overview

PROPOSED DATA SOURCE — BACKBLAZE DRIVE STATS

Dataset overview

Dataset at a glance

- Quarterly drive statistics
- Daily drive-level observations
- SSD model information
- Failure indicators
- SMART attributes
- Infrastructure identifiers

Key attributes

- `date`
- `serial_number`
- `model`
- `failure`
- SMART fields
- `pod_id`, `vault_id`
- `cluster_id`, `datacenter`

Relevance to the project

Dataset information	Analysis
SSD model and failure indicator	Model-wise failure analysis
Pod, vault and cluster identifiers	Infrastructure-wise analysis
SMART attributes	Drive health characteristics
Date and serial number	Historical drive-level tracking

Data source: Backblaze Drive Stats – official dataset

Note: SSD-specific failure records and infrastructure metadata coverage will be audited before finalising the experimental dataset. Backblaze's overall fleet statistics should not be interpreted as SSD-only statistics.

Expected Outcomes

- **Failure Pattern Analysis:** Identify failure concentrations across SSD models and infrastructure groups.
- **Maintenance Prioritisation:** Generate ranked maintenance lists based on historical failure distributions.
- **Budget-Based Evaluation:** Compare random, model-aware, infrastructure-aware and multi-factor selection strategies under different maintenance budgets.
- **Performance Assessment:** Evaluate strategies using failure coverage, selection precision and improvement over random selection.
- **Decision Support Dashboard:** Visualise failure trends, maintenance priorities, strategy comparisons and resource utilisation.

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Thank You